

Indirect access — using intelligence to make a virtue out of a necessity

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Indirect access using intelligent CPE means that customer dialling habits do not change when they switch to a second operator. This is an essential technology for second operators, in order to reach highly profitable mass-market customers and this paper describes some of the important features they should consider when designing such a service. In particular it highlights the need to consider complete service design rather than just equipment price. Remotely managed intelligent smartboxes and routers can support a range of alternative telco operation modes, such as switchless resellers and aggregators, and the use of indirect access to complement targeted access build is mentioned. Collaborative work on vendor and design analysis is reviewed and the point is made that remote management is a key ingredient that should be developed as a valuable marketing tool. This technology could support mass-market packet or Internet-based voice services and some operating models can be readily shifted to such a network without the customer needing to know. There are great similarities between dial up Internet service propositions and indirect access so the two services can be converged. It is stressed that indirect access is not a 'fallback' for direct access, but a true value add service in itself.

1. Introduction

Voice services were deregulated in most European countries from 1 January 1998 and, to seize this opportunity, second operators are formulating indirect access product-marketing plans. This paper will first outline how indirect access has developed and then review how this will fit in with service features that a second operator may wish to deploy. The underlying hardware will be described and a set of key equipment requirements and specifications outlined.

2. Indirect access service requirements

Indirect access involves delivering voice services via interconnect agreements. In the UK, the first competitors were 'facilities-based' telcos who own or lease switches and transport infrastructure. To reach their network, a user dials a carrier access code and then adds the normal routing digits, sometimes with a second dial tone or prompt message between the two. Routers, or 'smartboxes', make life easier and sit between an ordinary telephone and the socket.

For business users, private branch exchanges (PBXs) enforce specific traffic routings for outgoing calls, or remotely managed stand-intelligent 'least-cost routers' can be used. From an operator's perspective, smartboxes or PBX routers capture and route profitable customer traffic and discard the rest.

Figure 1 shows how a simple second operator, indirect-access service works. Previously, the customer had to remember to prefix the destination number if they wanted to route a call to the second operator. With intelligent routers, only the destination number is dialled by the customer, and the router decides whether to prefix the number with the second operator carrier select code or not, and thus route the call over its own core network, rather than that of the PTT. This allows 'screening' of calls to be made, and selective routing to take place. Service requirements can be stated in simple terms as follows:

- **for a customer** — customers want a service that is identical to, yet cheaper than, that from the incumbent PTT, and should not have to change their dialling habits,
- **for a second operator** — the disadvantages of a cheaper service using access via the PTT network should be kept to a minimum and opportunities provided by remotely managed intelligent customer premises equipment (CPE) should be exploited.

Figure 2 shows the spectrum of direct and indirect access services, ranging from residential (RES) through single line business (SLB), to small-to-medium enterprise (SME) and multinational/corporate (MNC) businesses.

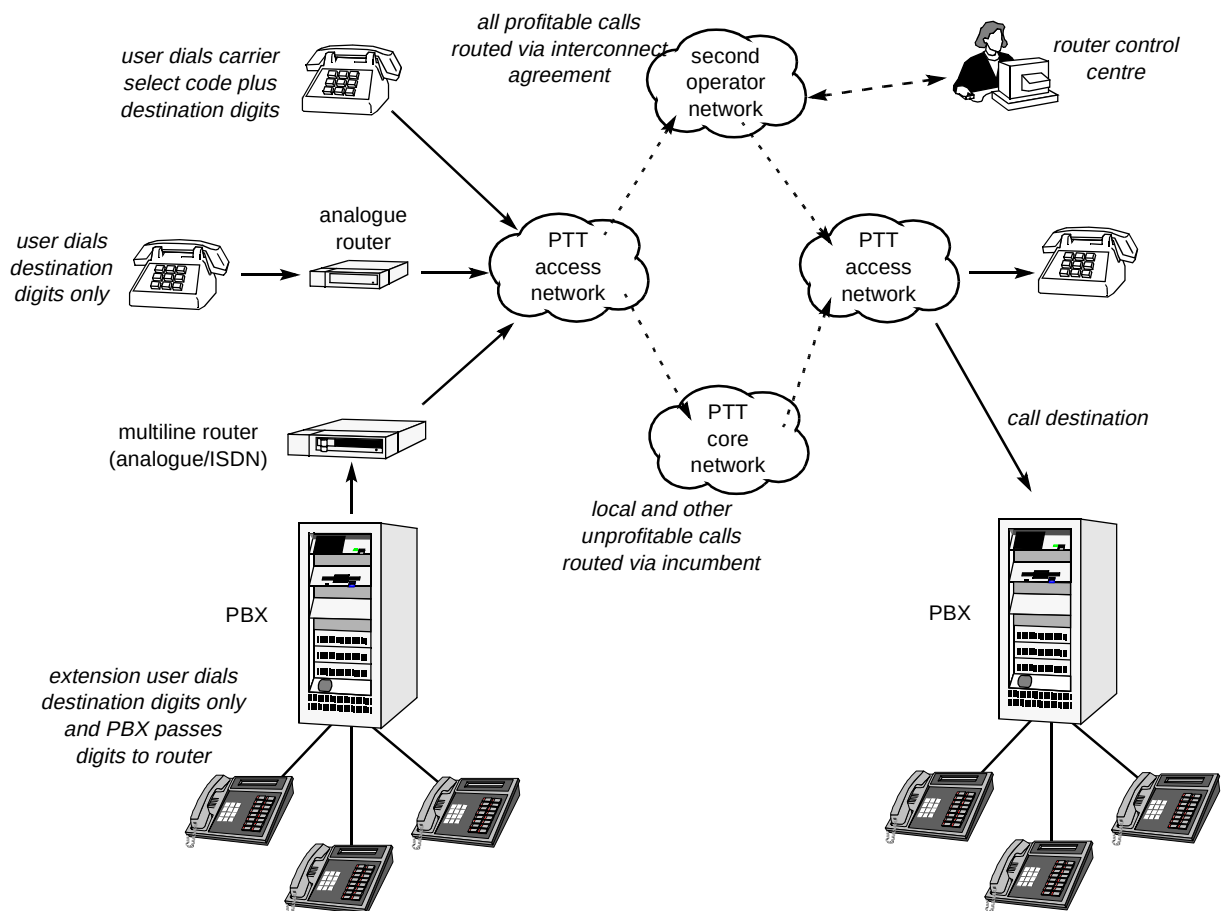


Fig 1 Indirect access.

Customers have a wide choice of competing operators for long-distance and international calls while paying the incumbent's line rental and local call charges. Indirect access has a major part to play for second operators, as it extends customer reach economically to small, scattered business and residential sites. Virtually all MNC sites are worth considering for developing into direct access — leased lines, fibre, wireless, dark copper, etc.

In densely populated areas, targeted fibre access build (TAB) looks promising for all customers, including residential ones. In sparse areas, only high-spending

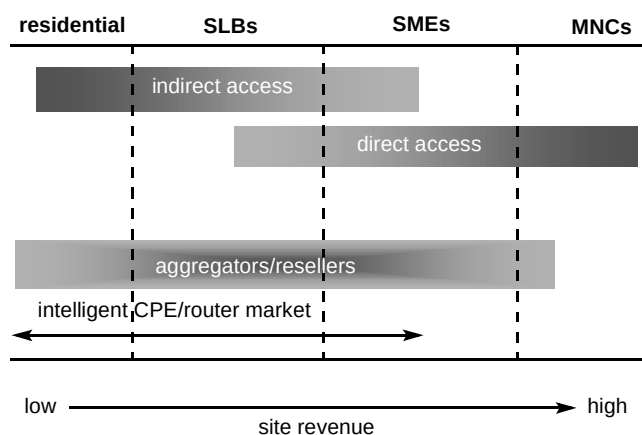


Fig 2 Indirect and direct access market boundaries.

residential sites such as small office/home office (SOHO) or ones making many international calls are really desirable for an indirect access service. The biggest challenge is being able to identify and attract these people while discouraging unprofitable customers. One way of widening the target market is a reduced-cost, 'no frills' service, paid for with a prepaid card (so there is no billing overhead), and poorer speech quality (adequate for customers benefiting from cheap international calls). These kinds of operator have been particularly successful with ethnic communities calling overseas relatives. Residential smartboxes can significantly reduce residential customer churn, and ideally should be given away free — if they are cheap enough. This also allows customer data mining to be enabled. For business customers, the distinction between SLB and residential SOHO customers may be small. SMEs are of considerable interest and most of these have some form of a PBX with typically six lines, together with dedicated fax lines. Larger sites may benefit from multiline routers which support direct lines to heavily used destinations and indirect access, to the rest. As mentioned, most MNC sites will eventually be served with direct access, but there are advantages in using indirect access routers to overflow traffic via the PSTN during busy hours.

One of the effects of this technology is that it supports registered site access (RSA), which in turn can blur the

boundaries between market segments. This occurs where customers have domestic or international virtual private networks (VPNs). A typical RSA site might be a motor car distributor, local authority or chain of estate agents where registered sites can include small offices or residential sites for remote teleworking. Calling line identification (CLI) is used to verify the status of a particular site. RSA tends to confuse the convenient boundaries between the different market segments as a customer may have significant SOHO and SLB sites while remaining a corporate client.

3. Entrant operators

Figure 3 shows that indirect access can support a wide range of operator types. As mentioned, a full telco includes direct (and possibly indirect) access whereas all the others rely on indirect access. For licensing purposes, a switch-based reseller may still be considered a facilities-based telco, even if the network is leased. The switchless reseller is a simple market entry model to sell cheap national/international calls via indirect access, where access to carriers is by carrier-select codes which can be programmed into the smartboxes and routers. The carrier choice will depend on destination. Callback operators are similar operations except that they exploit accounting rate and tariff inequalities, and are particularly active in closed markets. Aggregators take the dealer/agent model much further, acting as telecommunications service brokers, buying wholesale minutes or capacity from several carriers. Although in Fig 3 they are shown as simply a broker, with new technology they can look more like a switchless reseller, billing the customer from call-detail records downloaded from remotely managed routers. From a

customer's perspective, aggregators are interesting because they can proactively manage least-cost routing of the customer's traffic, offering a similar service to a telecommunications manager in a smaller company. There are signs of a move towards exploiting the developing switched minutes futures market in a true broker model and there must be great opportunities to do this profitably.

The operation modes described above align with the idea of the split in the industry between 'ServiceCo' and 'NetCo'. ServiceCo concentrates on establishing a market presence as a value added operator with customer acquisition, customer service and in some cases billing. It provides a simple market-entry model to sell cheap national/international calls via indirect access using wholesale capacity from NetCo. ServiceCo should always aim to maximise profits by least-cost routing. The advantage of the ServiceCo/NetCo split is that the ServiceCo may be able to exploit intelligent CPE in a way that competes more effectively with a telco who has not separated retail and wholesale operations. There may also be a situation where ServiceCos are competing against a telco or another operator for retail customers, while at the same time buying wholesale services from them. This is largely a branding issue.

4. Router hardware

Figure 4 shows the technology boundaries between analogue and digital routers. This boundary will vary from country to country with considerable overlap.

The underlying hardware for all smartboxes or routers is similar to a PC's architecture but the application will

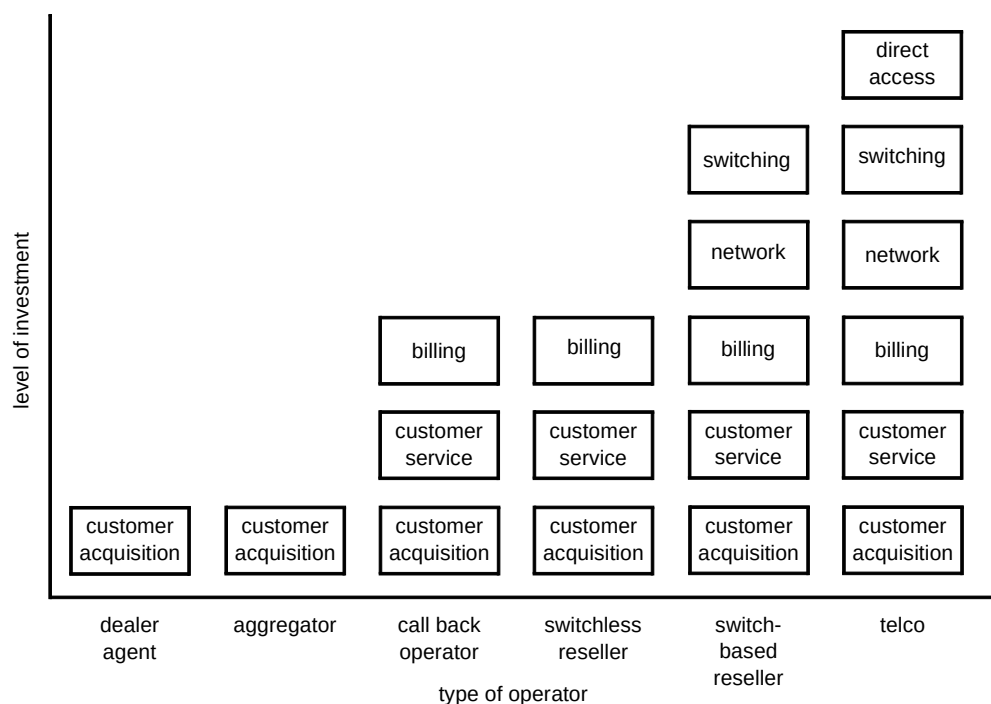


Fig 3 Different types of telco.

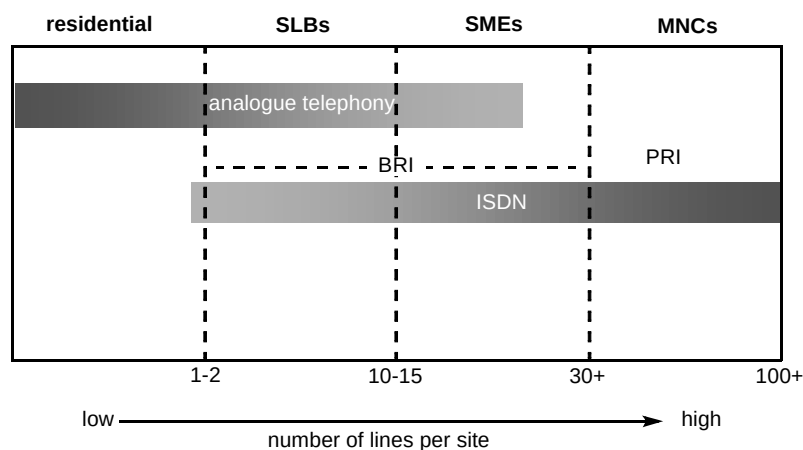


Fig 4 Indirect access technology boundaries.

significantly affect the router specification — a single line analogue unit is a totally different kind of product to a larger multiline unit. In most countries residential telephony is still analogue although in some countries SOHO customers may choose to install basic rate ISDN (BRI) which will require a terminal adapter. At the other end of the scale, virtually all large corporate sites will have digital access, with indirect access via primary rate (30+D) ISDN (PRI). Between the two a mixture of multiline analogue and BRI lines are likely to be used.

5. Router specifications

For RES and SLB applications, single line analogue units are available; they tend to be priced on volumes and functionality, and are designed to be mailed to the end user. The best examples are software programmable, remotely managed and line powered. Fully integrated intelligent telephones are likely to appear soon. Boxes are usually preprogrammed before delivery and can be updated either locally or remotely. For business applications, routers can be mounted on a wall near the PBX and installed by a technician with similar background and skills as a PBX maintainer. Multiline analogue units and ISDN BRI units are also available, and are usually priced on a per-line basis. PRI PBX boxes are based on PC technology, with Pentium processors and proprietary line cards that provide additional functionality such as 'any-to-any' protocol conversion for PBX network feature transparency. To develop a better understanding of router technology, detailed specifications have been developed, the more important of which are shown in Table 1.

Modularity is desirable so that customers only have to pay for the hardware they need to match the number of lines. Residential units should be small and line powering is very attractive. Remote management is essential and, to avoid fraud, the unit should call home to the control centre, rather than the opposite, which is vulnerable to unauthorised reprogramming or 'slamming'. On the whole, JVs are proposing to use routers to direct customer traffic to partners' networks, but most have also recognised the

advantages of using them as remote traffic pattern or route information recording devices. The key to success is the richness of the customer statistics that can be logged remotely. Time stamping with a real-time clock is also useful. In-country approval is essential and some countries such as Sweden accept BABT approval, whereas others, notably Switzerland and France, require additional testing which tends to be a means of discouraging non-domestic suppliers. Units should be transparent to incoming PSTN calls with minimal effect on post-dialling delay and quality of calls. In addition, routers for PRI ISDN customers should work as a protocol converter and support a range of subscriber signalling interfaces.

6. Operations and back office requirements

A back office organisation is necessary, and how the control centres handle indirect customers is an important part of the overall service design. Below is a list of the type of task to be undertaken:

- customer help-desk facilities,
- support for new customer installations by triggering router programming, despatch and installation procedures as they occur,
- recording location, software configuration, build state and software release for every box,
- investigation of automatic alarms and initiation of remedial action,
- processing and reviewing all statistics up-loaded from boxes and preparing required performance/management reports for carrier and customer,
- customer billing facilities,
- periodic review of all operating procedures and introduction of improvements,
- periodic review of in-country and international tariffs with advice to management on action required,
- preparation of download files for each box.

Table 1 Router technology overview.

Feature	Mass market (RES/SLB)	Small-to-medium enterprise	Corporate	
Number of lines	Single	2—8 lines, preferably modular and chainable	Single or multiple PRI interfaces	
Size	Small — can be posted	Small/medium — similar to ISDN terminal adapter	Medium — similar to rack mounted PC	
Installation	Plug and play for customer installation	Easy installation by technician	Trained installer	
New customer registration	Automatic network log-on			
Transparency	All incoming calls			
Post-dialling delay	Minimal (< 1 sec additional to network — or manually dialling carrier codes)			
Compatibility	Voice, fax and data calls			
Internal modem	Desirable	Essential	Uses ISDN signalling	
Management	Calls network management system, remote parameter configuration including carrier selection codes.	Full local and remote parameter configuration including carrier selection codes, statistics gathering and real-time clock.		
Management platform	Single windows GUI-based platform for analogue and digital units			
Routeing management	Support abbreviated/short code dialling Support VPN and customer dial/numbering plans			
Time of day routing	Desirable if real-time clock available	Time of day/service selection using real-time clock		
Remote configuration	Support remote configuration of parameters, including carrier selection codes	Comprehensive remote configuration of parameters, including carrier selection codes		
Congestion or route failure behaviour	Fall back to defined alternate carrier choice			
Customer statistics gathering	Single peg counters of call details downloaded and remotely configured in the unit.	Automatic uploading/downloading of call records to management system at preprogrammed intervals with time stamping to allow customer profiles to be established. Can include incoming call logging — time to answer and duration.		
Configuration management	Internal downloadable log of hardware/software configuration, version and upgrades			
Fraud protection	Support remote inventory control with electronic serial number and software checks			
Fault handling	Fallback to PSTN and ‘call home sick’. Local log files and routing tables kept intact. User notified if unit not functioning correctly.			
Power	Line powered but still supports high usage customers, fax machines or several extensions.	Mains power acceptable. Larger units can use PBX battery if required.		
On-line time for reprogramming	< 6 seconds	< 2 seconds		
Approval	In-country approval essential	In-country approval essential, pan-European approval acceptable for ISDN units		

The organisation will need skills to collect fault reports, provide analysis and determine corrective action on both hardware and software. Assuming successful market penetration the organisation will need to be able to manage several hundred thousand boxes or more. Information collected and stored into the remote centre database provides a rich source for data mining exercises and research, essential in formulating plans for future expansion.

7. Service development using indirect access

Indirect access service propositions need to be developed in a way that exploits the mix of all access technologies to extend customer reach. A good example is the capabilities of smartboxes and routers to gather customer calling patterns that include non-routed calls. This may be used as the precursor to an access build exercise as well as being a good way of reaching customers on the edge of the build zone. It has been mentioned that the service proposition should include excellent customer handling, and for this reason, it is worth considering outsourcing indirect access service management, at least in the short term. Several smartbox/router suppliers offer this.

In the longer term, remote management should be developed as a valuable marketing tool by the partners, particularly as the service proposition matures. A good example is the ability to 'build in' call-by-call service performance measurement capability. For certain markets it may be appropriate to move towards a ServiceCo/NetCo proposition, using indirect access as an opportunity to create strong branding and marketing identity from the outset. The important point with ServiceCo is an arm's length NetCo relationship, providing excellent customer handling while avoiding transport infrastructure investment and other traditional telco thinking. In the long term, ServiceCo and NetCo should be in the same group, both benefiting from shared market planning and technology information.

Throughout, this paper has emphasised the significance of remotely managed CPE placed on a customer's premises and how technology convergence can be used to develop new markets. This is called 'making virtue out of necessity' and there is a great opportunity for new ideas for service bundles. For instance, hotels could have a resident's telephone service linked with remotely managed intelligent vending machines alongside inventory control and automatic ordering of supplies.

8. Indirect access and packet-based core network transport

It is worth mentioning that indirect access can be used for voice transport over packet systems such as ATM, frame relay, or Internet. This is attractive because the idea of the integrated multi-service network, being a small step from intelligent CPE to a mass-market voice service over an existing frame relay or ATM network. The key to success is that some indirect access operating models can be shifted to a packet-based core network using a voice gateway without the customer needing to know about it — provided that the regulatory environment allows it. This is 'switchless' using virtual circuits via the packet network access device. Far-end delivery is the reverse. For a reasonable size country, it is thought that 20—25 of these gateways would be required. Figure 5 shows an example of such a service with a frame relay network.

There are also great similarities between dial-up Internet service propositions and indirect access. This means it should be possible to make the two services converge by converting to voice over Internet at the indirect access service gateway. The most interesting outcome of this convergence is where a fully integrated multiservice network is being developed. In this case the network will support direct customers with integrated voice and data services, probably as a corporate intranet for MNCs while at the same time providing indirect access voice and data

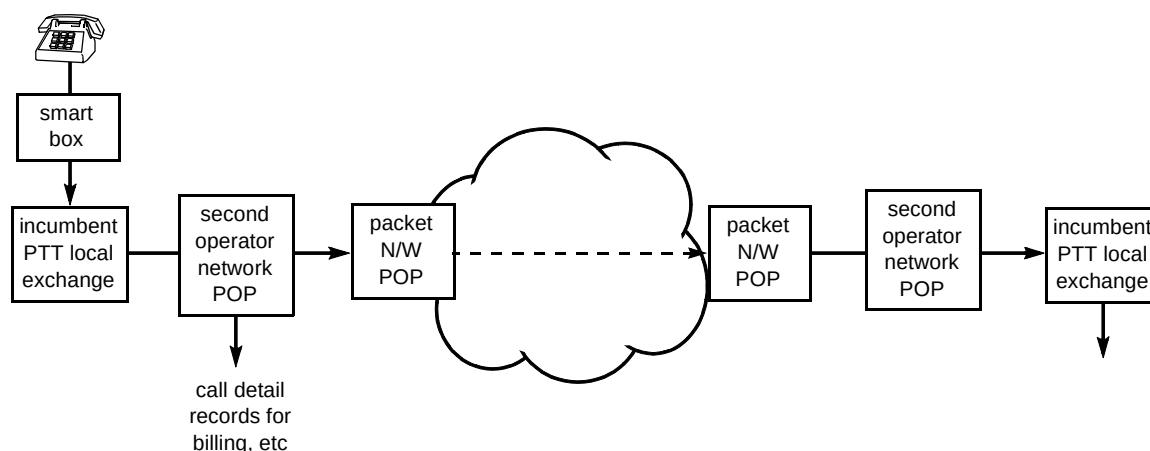


Fig 5 Indirect access voice services over a packet based core network.

Table 2 Indirect access and IP comparison.

Service feature	Indirect access	Dial-up Internet service provider
PoP access mechanism	Dials up pre-programmed number and awaits confirmation signal.	Dials up pre-programmed number and awaits confirmation signal.
Customer authentication	PIN and authorisation code. CLI used if available.	In most cases PIN and authorisation code. CLI beginning to be used for services such as BT Click.
Customer identification	Electronic serial number held in smartbox matched against CLI and customer details.	User name inserted by customer at log-on.
Billing	Prepayment cards or call detail records from switch or PoP.	Subscription and/or connect time charges.
Access charges	Incumbent bills local calls and line rental. Call fee includes second operator access charges.	Incumbent bills local calls and line rental — calls to service provider paid by user.
CPE ownership	Second operator	User
Characteristics of CPE	Cost-reduced custom PC architecture with integral modem in more expensive units.	'Dumb' modem linked to general-purpose PC running Windows-based applications.
Software features	Custom software licensed and maintained by second operator.	General-purpose browser software licensed and maintained by end user.
Service reliability	It is in the interests of the second operator to dimension the access to their network correctly for reliable call set-up. Once connection is set up, calls rarely drop out, even in busy periods.	Remote connection to ISP is no guarantee that user can complete server connection. Connections regularly drop out, particularly in busy times.

services for the mass markets. Table 2 compares some of the service features.

9. Conclusions

Indirect access is important for small, scattered sites where it is probably the only sensible way for a second operator to provide service to highly profitable mass-market customers. This paper has outlined the key features of router specifications, of which low cost, simple installation, remote management, and ability to gather call statistics are all important. Features of indirect access service surround have been mentioned, particularly with respect to developing operations and target customers. Finally, but by no means least, it was stressed how indirect access can operate closely with other voice transport systems. Its importance lies not only with routing calls to a second operator network, but with other voice services, such as voice over frame relay, and voice over Internet. Indirect access is not a 'fallback' for direct access, but a true value add service in itself.

Acknowledgement

Figure 3 is reproduced with permission from the Ovum report 'Resale and callback: opportunities and threats', published in 1998.



Colin Mallett currently holds design authority for indirect access on behalf of BT Europe's Technical Leadership Programme. He joined BT Laboratories in 1968 and has a BSc Electrical Engineering from Bath University and an MSc in Telecommunications from Essex University. He has worked in international communications for over 20 years, initially with undersea optical systems and more recently with the European Joint Ventures and Concert. Current activities involve managing a collaborative project on a conceptual JV with Telenor and work on the Alliance Partners' Strategic Agenda. These

are linked to use of competitor analysis, performance bench-marking, integration, second operator technology and ServiceCo concepts. He has published numerous papers, co-authored a book on high-speed logic, and been a guest editor of the IEEE Journal of Lightwave Technology.



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